

Lab: The Complete Active Inference Agent

Learning Goal: Build, run, and analyze the fully integrated Active Inference agent.

Part 1: Initialize the Complete Agent

```
# 2-state foraging environment
A_prior = np.ones((2, 2)) * 2.0
B_prior = np.ones((2, 2, 2)) * 2.0
C = np.array([3.0, -1.0])
D = np.array([0.5, 0.5])

agent = CompleteActiveInferenceAgent(A_prior, B_prior, C, D, policy_len=1)
print(f"Policies: {agent.policies}")
print(f"Initial A:\n{agent.A.round(3)}")
print(f"Initial habit entropy: {-(agent.policy_prior * np.log(agent.polic
```

Write your response here

Part 2: Run 100 Steps

```
true_A = np.array([[0.9, 0.1], [0.1, 0.9]])
true_B = np.zeros((2, 2, 2))
true_B[:, :, 0] = np.eye(2)
true_B[:, :, 1] = np.array([[0, 1], [1, 0]])

env = SimpleEnvironment(true_A, true_B)
agent = run_full_simulation(agent, env, num_steps=100)
```

Analyze: Did the agent learn the environment? Did habits form?

Write your response here

Part 3: Analyze the Four Subsystems

After 100 steps, examine each subsystem:

1. **Perception:** Compare learned A to true A — how close?
2. **Action:** What is the distribution over policies in the last 10 steps?
3. **Learning:** Plot the learning rate trajectory
4. **Habits:** Print the habit strength and policy prior

```
print(f"Learned A:\n{agent.A.round(3)}")
print(f"True A:\n{true_A}")
print(f"Final policy prior: {agent.policy_prior.round(3)}")
print(f"Habit strength: {agent.habit_strength.round(1)}")
```

Write your response here

Part 4: Goal-Directed vs. Habitual Transition

Exercise: Compare early steps (1-10) vs. late steps (90-100):

- How do policy probabilities differ?
- Is the agent more "habitual" by step 100?
- What is the role of EFE vs. policy prior at each phase?

Write your response here

Part 5: Capstone Reflection

In 200 words, reflect on the entire Implementation unit: You've built an Active Inference agent from scratch — data structures, inference, action selection, learning, communication, and planning. What are the most important insights from translating theory into code? What would you add or change?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Complete initialization	Full agent setup
2	Extended simulation	100-step run
3	Subsystem analysis	Perception + Action + Learning + Habits
4	Behavioral transition	Goal-directed → Habitual
5	Capstone reflection	Theory ↔ Implementation synthesis